

IRA-26-1466
DISTRIBUTION STATEMENT A

Description:

Perform two Close Interval and Direct Voltage Current Gradient Surveys at Arnold AFB, TENNESSEE. Approximately 19,938 linear feet of UG 161kV oil filled pipeline (system 1) and 3,500 linear feet of UG fuel pipeline (system 2).

Qualifications:

All corrosion assessment and repair work shall be performed under the supervision of a NACE Certified Corrosion Specialist. NACE/AMPP certified (Level 4), corrosion control engineer, and testing personnel shall be supervised by minimum of NACE/AMPP Level 2, Cathodic Protection Technician certification. Submit evidence of qualifications for personnel performing this work.

Indirect Assessment:

Perform Close Interval Potential Survey (CIS) for 161kV Underground Pipeline Distribution system IAW with NACE/AMPP standards and industry best practices.

Perform Direct Current Voltage Gradient (DCVG) survey for 161kV underground Pipeline Distribution system IAW with NACE/AMPP standards and industry best practices.

Perform CIS for UG Fuel Lines IAW NACE/AMPP standards and industry best practices.

Perform DCVG for UG Fuel Lines IAW NACE/AMPP standards and industry's best practices.

Submit results of each survey (CIS AND DCVG) in report form encompassing engineering analysis and recommendations for technical, logistical, and life cycle support.

Direct Assessment:

Direct Assessment shall be based on NACE SP0502 and shall include all excavations and tests that are deemed necessary to confirm and classify the general condition of the pipe. At the excavation sites, document the condition of the pipe coating, pipe, and backfill around the pipe. Perform necessary repairs to the pipe, coating, and cathodic protection system using materials that are compatible with existing materials. Submit plans for repairs and product information. The Contractor shall not commence any work without Contractor approval. The approximated depths of pipe cables range from 5 feet to 13 feet.

Final Assessment Report: Prepare a written report containing the final documentation from the direct assessments. Report shall include a list of repairs what was performed and GPS coordinates showing where repairs were performed. Include assessment of root cause. Provide recommendations for additional repairs and/or improvements to the

IRA-26-1466
DISTRIBUTION STATEMENT A

cathodic protection system for the overall health of the pipe. Include breakdown of all costs for repairs in identified locations. Include predictions on the integrity of the remaining, unexamined pipe. Include inspection and preventive maintenance cycle recommendations of the pipe and cathodic protection system. Provide any as-built drawings in accordance with Appendix A.2 – Submittal Procedure. Submit final report for approval to the Government.

Drivable Markers for UG:

Contractor shall install red fiberglass drivable markers every 200' along each underground cable, except in parking lots, roads, and driveways. The top of each marker shall have "WARNING UG 161kV CABLE" printed on durable vinyl film with protective overlay for outdoor durability.

History:

The approximate depths of the buried 161kV pipe-type cables are a minimum of 5 feet to a maximum of 13 feet. There are multiple crossings of other underground utilities such as raw water and natural gas.

The pipe coating contains asbestos. Coordinate with Government and dispose of in accordance with Appendix A.4 – Asbestos Removal.

The electrical underground distribution system was built in the 1950's, with no indication or records these lines were initially cathodically protected. The 161kV Pipeline CP System was originally designed and built in 1975 with a single, deep well anode bed 10-inch diameter x 260 feet in depth. The system, as originally configured, was designed to be isolated from the copper grounding systems at the various power substations.

In 2009 a survey determined the 161kV pipeline system was shorted to the UG fuel pipeline system through a common grounding system. Survey also indicated the deep ground anode bed had failed as well as noting several failed CP system rectifiers.

In 2012 nine new impressed-current cathodic protection (ICCP) systems were installed, one for each 161kV substation. The new CP systems consist of mixed-metal oxide (MMO) anode beds, negative cable wiring and Universal ASAI rectifiers. During the initial survey, after installation of this new system, it was noted that significantly more positive ON and OFF potentials were observed within the substation fencing. This is due to the copper grounding grid inside each station's perimeter, requiring special considerations to be employed to obtain accurate survey results of the oil filled pipelines.

In 2017 a CIS and DCVG survey was completed on the 161kV and UG Fuel pipelines which indicated three areas as Severe- Immediate Action items; twenty-nine areas classified as Moderate - Scheduled Action items; and thirty-three areas as Minor-Monitored Action items. Refer to NACE/AMPP Standard SP0502-2010 Section 5.2 Paragraph 5.2.2 regarding the indications of Severe, Moderate, and Minor per this subparagraph. The 2017 report is provided in Appendix E.

IRA-26-1466
DISTRIBUTION STATEMENT A

CIRCUIT	CIRCUIT LENGTH FEET*	CABLE SIZE	SPLICED	INSTALLED YEAR
UG 91	1885	1500 MCM	YES	1953
UG 92	2943	1500 MCM	YES	1953
UG 93	1090	1500 MCM	NO	1953
UG 94	805	1500 MCM	NO	1953
UG 95	1221	1500 MCM	NO	1953
UG 96	390	1500 MCM	NO	1957
UG 97	902	1500 MCM	NO	1957
UG 98	3500	1500 MCM	YES	1959
UG 81	2532	2500 MCM	YES	1982
UG 82	2593	2500 MCM	YES	1982
UG 83	835	2000 MCM	YES	1982
UG 84	1242	2000 MCM	YES	1982

Table 1- List of Underground 161kV Cable Pipelines